AISHWARYA NAYAK

Senior AI Engineer | Applied AI | Backend Systems

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SUMMARY

Senior Software Engineer and Applied AI Engineer with 5 years of experience delivering distributed systems, cloud-native services, and AI-driven applications across USA and India teams. Backed by an MS in Data Science from the University at Buffalo (SUNY), with expertise spanning backend engineering, applied AI, frontier-model evaluation, and software-engineering benchmarks. Experienced in building scalable APIs, reproducible testing frameworks, and reliable AI systems while combining strong software engineering fundamentals with modern machine learning practices.

SKILLS

AI & Machine Learning: Large Language Models (LLMs), AI Evaluation, AI Benchmarking, Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, OpenAI APIs, Gemini, Model Evaluation, Error Analysis, Scientific Integrity Evaluation, Human-in-the-Loop Evaluation, Computer Vision
Languages: Python, Java, Scala, JavaScript, TypeScript, SQL
Backend & Distributed Systems: Microservices, REST APIs, FastAPI, Apache Kafka, Event-Driven Architecture, Play Framework, Lagom, Knowledge Graphs
Cloud & Infrastructure: Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD, AWS, GCP, Vertex AI
Databases & Search: MySQL, Apache Solr, ChromaDB, Sentence Transformers
Developer Tools: Git, Linux, n8n, MCP, LaTeX, KaTeX

PROFESSIONAL EXPERIENCE

Handshake AI — AI Engineer / Software Engineer (AI Evaluation)
Remote (USA & India Projects) | Sep 2025 – Present

Worked across multiple AI research and evaluation initiatives involving frontier LLM reasoning, coding agents, multimodal systems, scientific integrity, and software-engineering benchmark development.

Software Engineering Benchmark Development

- Authored SWE-bench-style benchmark tasks for evaluating frontier coding models on real-world open-source repositories.
- Designed realistic bug fixes, feature enhancements, API migrations, and edge-case scenarios across large unfamiliar codebases.
- Developed deterministic Fail-to-Pass (F2P) and Pass-to-Pass (P2P) validation suites to ensure behavioral correctness while remaining implementation-agnostic.
- Built reproducible Docker-based execution environments and automated benchmark validation workflows using GitHub Actions and custom test runners.
- Engineered benchmark artifacts including metadata schemas, debugging hints, reference implementations, and evaluation instructions aligned with single-turn coding-agent assessment protocols.
- Performed repository analysis, pull-request reviews, and test coverage validation across Python, JavaScript, and TypeScript projects.

LLM Code Evaluation

- Evaluated model-generated code across Python, JavaScript, C++, R, and additional languages using execution-based quality frameworks.
- Executed and validated generated programs within isolated Docker environments to identify runtime failures, hallucinations, and hidden logic inconsistencies.
- Generated reproducible evaluation evidence using execution traces, terminal logs, outputs, and screenshots.
- Collaborated asynchronously with engineers and product teams while adapting to evolving evaluation policies and tooling environments.

Scientific Integrity Evaluation

- Audited AI agent traces for scientific reproducibility and methodological correctness across machine learning experiments.
- Detected data leakage, metric hacking, hidden evaluation manipulations, test-set access violations, and plan-code inconsistencies.
- Produced structured integrity assessments with evidence-backed justifications and log-based citations.

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Prompt Engineering & Reasoning Evaluation

- Designed 200+ challenging reasoning tasks for multimodal and text-only frontier models using domain-specific evaluation methodologies.
- Produced human-verifiable solutions and enforced strict answer schemas using LaTeX and KaTeX.
- Reviewed prompts using structured rubrics to identify ambiguity, plagiarism, and reasoning inconsistencies.

Multimodal Dataset Collection

- Contributed to multimodal dataset generation initiatives under strict privacy and quality requirements.
- Performed quality assurance checks to ensure consistency for downstream computer vision and navigation systems.

Reality AI Lab — Generative AI Engineer
Buffalo, New York, USA | Mar 2025 – Sep 2025

- Developed a retrieval-augmented AI teaching assistant enabling interactive academic dataset querying using Python, LangChain, and OpenAI APIs.
- Built scalable ingestion and preprocessing pipelines supporting long-context LLM applications, reducing input errors by 15%.
- Deployed models using GCP Vertex AI and containerized services, automating model training and production inference workflows.
- Integrated logging, evaluation, and monitoring frameworks to improve reliability and accelerate root-cause analysis.
- Collaborated with backend engineers to design modular APIs supporting knowledge graph updates and prompt orchestration workflows.
- Contributed to the end-to-end lifecycle of AI applications spanning data ingestion, model deployment, inference, and observability.

Mocapay — Software Engineer, Backend
Delhi, India | Aug 2021 – Jun 2023

- Developed and maintained reactive Scala microservices supporting fintech products generating over $10M in annual revenue.
- Designed and implemented RESTful APIs for large-scale transaction processing systems, improving reliability and customer experience.
- Integrated Apache Kafka for asynchronous communication and event-driven workflows, enhancing real-time transaction capabilities and system resilience.
- Streamlined CI/CD pipelines using Jenkins and Docker, reducing deployment time by 30%.
- Enhanced platform robustness and security in accordance with Vanta recommendations, contributing to SOC 2 compliance.
- Improved search capabilities by integrating Apache Solr, increasing search efficiency by 25%.
- Led code reviews and promoted software engineering best practices to improve maintainability and code quality.

Tata Consultancy Services (TCS) — Software Engineer, Backend
Indore, India | Jan 2020 – Jul 2021

- Led development of a finance platform supporting management of over 1,000 contracts and improving operational efficiency.
- Designed backend services and RESTful APIs for contract creation, validation, and payment calculations.
- Managed cloud deployments on AWS and GCP, reducing infrastructure costs by 15%.
- Optimized SQL queries and improved database performance by 20%.
- Implemented security controls and role-based access mechanisms using Fortify and SonarQube, reducing vulnerabilities by 40%.
- Automated testing and deployment pipelines using Jenkins, significantly shortening release cycles.
- Collaborated with cross-functional teams during sprint planning, backlog refinement, and product roadmap discussions.

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SELECTED PROJECTS

Enterprise Knowledge Copilot | RAG Knowledge Assistant [GitHub]

Tech Stack: Python, LangChain, Vector Databases, OpenAI APIs, FastAPI

- Built an enterprise-grade Retrieval-Augmented Generation (RAG) platform for contextual question answering and knowledge discovery across organizational documents.
- Designed scalable ingestion and retrieval pipelines supporting structured and unstructured data sources.
- Implemented semantic search and source-grounded response generation to improve answer quality and reduce hallucinations.
- Developed modular APIs and extensible workflows supporting future tool integrations and agent-based interactions.
- Enabled efficient knowledge access and productivity improvements through context-aware conversational interfaces.

CodeSage | Multi-Agent Repository Intelligence Platform [GitHub]

Tech Stack: Python, LangGraph, Pydantic, GitHub APIs

- Developed a LangGraph-based multi-agent system for automated repository analysis and architecture understanding.
- Built specialized agents for dependency inspection, quality assessment, security analysis, and documentation generation.
- Implemented shared-state orchestration and structured outputs using Pydantic to improve reliability and maintainability.
- Automated report generation workflows to accelerate onboarding and analysis of unfamiliar codebases.
- Designed extensible agent workflows for evaluating repositories and enhancing developer productivity.

FlowForge AI | AI Workflow Automation Platform [GitHub]

Tech Stack: n8n, Python, APIs, LLM Integrations

- Developed an AI-driven workflow automation platform enabling orchestration of multi-step tasks through modular and event-driven pipelines.
- Built reusable workflows integrating external APIs and large language models to automate repetitive processes and improve productivity.
- Designed extensible automation patterns supporting tool interactions, conditional execution, and multi-stage task coordination.
- Leveraged low-code orchestration principles to simplify integrations and accelerate workflow development.
- Enhanced operational efficiency through reusable components and scalable workflow design.

EDUCATION

Master of Science (MS) in Data Science — Aug 2023 - Feb 2025
University at Buffalo (State University of New York, SUNY Buffalo) — New York, US

Bachelor of Technology (B.Tech) in Electronics and Telecom Engineering — Jul 2015 - Apr 2019
Symbiosis International University — Pune, IN